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To: Selection Committee
Faculty of Electrical Engineering, Mathematics and Computer Science
University of Twente, The Netherlands

Subject: Application for PhD Position on Explainable DeepFake Detection Using Foundation Models

Dear Dr Spreeuwers and Members of the Selection Committee,

I am writing to apply for the PhD position on Explainable DeepFake Detection Using Foundation Models at the University of Twente. I hold an MSc in Data Science with Advanced Research from the University of Hertfordshire, which I completed with Distinction, and a BSc in Computer Software Engineering from Foundation University Islamabad. I am keen to continue my academic journey through doctoral research that applies artificial intelligence to a real-world problem, and I find the challenge of detecting manipulated images, videos and audio particularly meaningful.

My master's research focused on large language models and parameter-efficient fine-tuning methods, including QLoRA and IA^3. Through this work, I explored how large models can be adapted to specific tasks while working within limited computational resources. It also encouraged me to think carefully about model performance, evaluation and the practical limitations of AI systems. Although my dissertation was not specifically about DeepFake detection, it gave me a foundation in applied machine learning and strengthened my interest in exploring how modern AI models can be used to address different technical challenges.

What particularly interests me about this position is the emphasis on explainability. Identifying manipulated media is important, but in areas such as journalism and forensic investigation, simply labelling a video as genuine or fake may not be enough. Researchers also need to understand what evidence supports a model's decision, where inconsistencies appear and how reliable that conclusion is. I would like to investigate how foundation models and other AI techniques can help detect manipulation while producing explanations that people can assess and understand.

Alongside my academic background, I have professional experience in software development, machine learning and AI, as well as experience teaching programming and AI-related subjects. These roles have helped me develop a practical approach to technical problems and strengthened my ability to communicate complex ideas clearly. I enjoy learning how systems work, examining their limitations and finding ways to improve them. I believe these qualities would be valuable in a research project that combines modelling, development and experimental validation.

I understand that this position requires a strong interest in computer vision, pattern recognition and careful analysis of experimental results. I am motivated to deepen my knowledge in these areas and apply my existing machine learning background to the specific challenges of audio-visual DeepFake detection. I am also attracted by the opportunity to collaborate with researchers from different institutions through the DEFORM project. Working in an international research environment would allow me to learn from different perspectives while contributing my own technical experience and commitment.

I would welcome the opportunity to develop as a researcher at the University of Twente and contribute to research that improves the reliability and explainability of AI-based media analysis. Thank you for considering my application. I would be pleased to discuss my background, research interests and motivation further.

Yours sincerely,
Mirza Muhammad Jamal Nasir